

Acceleration

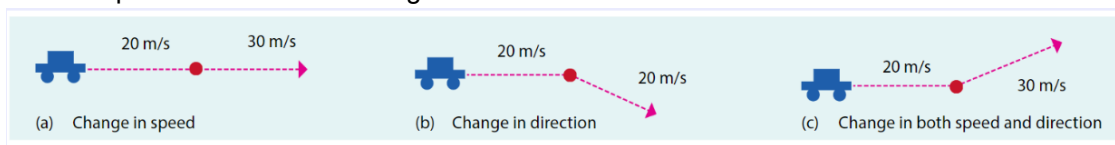
- object accelerates when velocity changes

Definition: acceleration is the rate of change of velocity per unit time.

#PhysicsDefinitions

Object undergoes acceleration when:

- speed or direction changes
- both its speed and direction change



SI unit: m/s^2

Formula

$$\text{Acceleration} = \frac{\text{change of velocity}}{\text{time taken}}$$

$$\text{uniform acceleration, } a = \frac{\Delta v}{\Delta t} = \frac{v - u}{t_v - t_u}$$

change v = change in velocity (m/s)

change t = time interval between t_v and t_u (s)

V = final velocity

U = initial velocity

t_v = time at which object is at final velocity (s)

t_u = time at which object is at initial velocity

Acceleration $\Rightarrow$ VECTOR $\Rightarrow$ can be negative

Uniform acceleration

- change (increase/decrease) in velocity of object for every unit of time = same
- Object undergoes constant/uniform acceleration

Definition: Uniform acceleration is the constant rate of change of velocity

#PhysicsDefinitions

Non uniform acceleration:

- object undergoes non uniform acceleration if change in velocity for every unit of time is NOT the same
-